

✓ Project Title: *Object Detection in Images*

Object detection is a fundamental computer vision task that involves identifying and localizing multiple objects within an image or video. It goes beyond simple image classification by not only recognizing what objects are present but also determining where they are located in the visual scene.

```
!git clone https://github.com/ultralytics/yolov5
```

```
%cd yolov5
```

```
!pip install -r requirements.txt
```

```
Requirement already satisfied: scipy>=1.4.1 in /usr/local/lib/python3.11/dist-packages (from -r requirements.txt (line 13)) (1.15.3)
Collecting thop>=0.1.1 (from -r requirements.txt (line 14))
  Downloading thop-0.1.1.post2209072238-py3-none-any.whl.metadata (2.7 kB)
Requirement already satisfied: torch>=1.8.0 in /usr/local/lib/python3.11/dist-packages (from -r requirements.txt (line 15)) (2.6.0+cu124)
Requirement already satisfied: torchvision>=0.9.0 in /usr/local/lib/python3.11/dist-packages (from -r requirements.txt (line 16)) (0.19.0)
Requirement already satisfied: tqdm>=4.66.3 in /usr/local/lib/python3.11/dist-packages (from -r requirements.txt (line 17)) (4.67.1)
Collecting ultralytics>=8.2.34 (from -r requirements.txt (line 18))
  Downloading ultralytics-8.3.144-py3-none-any.whl.metadata (37 kB)
Requirement already satisfied: pandas>=1.1.4 in /usr/local/lib/python3.11/dist-packages (from -r requirements.txt (line 27)) (2.2.2)
Requirement already satisfied: seaborn>=0.11.0 in /usr/local/lib/python3.11/dist-packages (from -r requirements.txt (line 28)) (0.13.2)
Requirement already satisfied: setuptools>=70.0.0 in /usr/local/lib/python3.11/dist-packages (from -r requirements.txt (line 42)) (75.8.2)
Requirement already satisfied: gitdb<5,>=4.0.1 in /usr/local/lib/python3.11/dist-packages (from gitpython>=3.1.30->-r requirements.txt (line 43)) (4.0.11)
Requirement already satisfied: contourpy>=1.0.1 in /usr/local/lib/python3.11/dist-packages (from matplotlib>=3.3->-r requirements.txt (line 44)) (1.1.1)
Requirement already satisfied: cycler>=0.10 in /usr/local/lib/python3.11/dist-packages (from matplotlib>=3.3->-r requirements.txt (line 44)) (0.12.1)
Requirement already satisfied: fonttools>=4.22.0 in /usr/local/lib/python3.11/dist-packages (from matplotlib>=3.3->-r requirements.txt (line 44)) (4.22.0)
Requirement already satisfied: kiwisolver>=1.3.1 in /usr/local/lib/python3.11/dist-packages (from matplotlib>=3.3->-r requirements.txt (line 44)) (1.4.5)
Requirement already satisfied: packaging>=20.0 in /usr/local/lib/python3.11/dist-packages (from matplotlib>=3.3->-r requirements.txt (line 44)) (24.1)
Requirement already satisfied: pyparsing>=2.3.1 in /usr/local/lib/python3.11/dist-packages (from matplotlib>=3.3->-r requirements.txt (line 44)) (3.1.4)
Requirement already satisfied: python-dateutil>=2.7 in /usr/local/lib/python3.11/dist-packages (from matplotlib>=3.3->-r requirements.txt (line 44)) (2.9.0)
Requirement already satisfied: charset-normalizer<4,>=2 in /usr/local/lib/python3.11/dist-packages (from requests>=2.32.2->-r requirements.txt (line 45)) (3.4.0)
Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.11/dist-packages (from requests>=2.32.2->-r requirements.txt (line 45)) (3.10)
Requirement already satisfied: urllib3<3,>=1.21.1 in /usr/local/lib/python3.11/dist-packages (from requests>=2.32.2->-r requirements.txt (line 45)) (2.2.3)
Requirement already satisfied: certifi>=2017.4.17 in /usr/local/lib/python3.11/dist-packages (from requests>=2.32.2->-r requirements.txt (line 45)) (2025.1.1)
Requirement already satisfied: filelock in /usr/local/lib/python3.11/dist-packages (from torch>=1.8.0->-r requirements.txt (line 15)) (3.16.1)
Requirement already satisfied: typing-extensions>=4.10.0 in /usr/local/lib/python3.11/dist-packages (from torch>=1.8.0->-r requirements.txt (line 15)) (4.12.0)
Requirement already satisfied: networkx in /usr/local/lib/python3.11/dist-packages (from torch>=1.8.0->-r requirements.txt (line 15)) (3.4.2)
Requirement already satisfied: jinja2 in /usr/local/lib/python3.11/dist-packages (from torch>=1.8.0->-r requirements.txt (line 15)) (3.1.4)
Requirement already satisfied: fsspec in /usr/local/lib/python3.11/dist-packages (from torch>=1.8.0->-r requirements.txt (line 15)) (2024.10.0)
Collecting nvidia-cuda-nvrtc-cu12==12.4.127 (from torch>=1.8.0->-r requirements.txt (line 15))
  Downloading nvidia_cuda_nvrtc_cu12-12.4.127-py3-none-manylinux2014_x86_64.whl.metadata (1.5 kB)
Collecting nvidia-cuda-runtime-cu12==12.4.127 (from torch>=1.8.0->-r requirements.txt (line 15))
  Downloading nvidia_cuda_runtime_cu12-12.4.127-py3-none-manylinux2014_x86_64.whl.metadata (1.5 kB)
Collecting nvidia-cuda-cupti-cu12==12.4.127 (from torch>=1.8.0->-r requirements.txt (line 15))
  Downloading nvidia_cuda_cupti_cu12-12.4.127-py3-none-manylinux2014_x86_64.whl.metadata (1.6 kB)
Collecting nvidia-cudnn-cu12==9.1.0.70 (from torch>=1.8.0->-r requirements.txt (line 15))
  Downloading nvidia_cudnn_cu12-9.1.0.70-py3-none-manylinux2014_x86_64.whl.metadata (1.6 kB)
Collecting nvidia-cublas-cu12==12.4.5.8 (from torch>=1.8.0->-r requirements.txt (line 15))
  Downloading nvidia_cublas_cu12-12.4.5.8-py3-none-manylinux2014_x86_64.whl.metadata (1.5 kB)
Collecting nvidia-cufft-cu12==11.2.1.3 (from torch>=1.8.0->-r requirements.txt (line 15))
  Downloading nvidia_cufft_cu12-11.2.1.3-py3-none-manylinux2014_x86_64.whl.metadata (1.5 kB)
Collecting nvidia-curand-cu12==10.3.5.147 (from torch>=1.8.0->-r requirements.txt (line 15))
  Downloading nvidia_curand_cu12-10.3.5.147-py3-none-manylinux2014_x86_64.whl.metadata (1.5 kB)
Collecting nvidia-cusolver-cu12==11.6.1.9 (from torch>=1.8.0->-r requirements.txt (line 15))
  Downloading nvidia_cusolver_cu12-11.6.1.9-py3-none-manylinux2014_x86_64.whl.metadata (1.6 kB)
Collecting nvidia-cusparselt-cu12==12.3.1.170 (from torch>=1.8.0->-r requirements.txt (line 15))
  Downloading nvidia_cusparselt_cu12-12.3.1.170-py3-none-manylinux2014_x86_64.whl.metadata (1.6 kB)
Requirement already satisfied: nvidia-cusparse-cu12==12.3.1.170 in /usr/local/lib/python3.11/dist-packages (from torch>=1.8.0->-r requirements.txt (line 15)) (12.3.1.170)
Requirement already satisfied: nvidia-nccl-cu12==2.21.5 in /usr/local/lib/python3.11/dist-packages (from torch>=1.8.0->-r requirements.txt (line 15)) (2.21.5)
Requirement already satisfied: nvidia-nvtx-cu12==12.4.127 in /usr/local/lib/python3.11/dist-packages (from torch>=1.8.0->-r requirements.txt (line 15)) (12.4.127)
Collecting nvidia-nvjitlink-cu12==12.4.127 (from torch>=1.8.0->-r requirements.txt (line 15))
  Downloading nvidia_nvjitlink_cu12-12.4.127-py3-none-manylinux2014_x86_64.whl.metadata (1.5 kB)
Requirement already satisfied: triton==3.2.0 in /usr/local/lib/python3.11/dist-packages (from torch>=1.8.0->-r requirements.txt (line 15)) (3.2.0)
Requirement already satisfied: sympy==1.13.1 in /usr/local/lib/python3.11/dist-packages (from torch>=1.8.0->-r requirements.txt (line 15)) (1.13.1)
Requirement already satisfied: mpmath<1.4,>=1.1.0 in /usr/local/lib/python3.11/dist-packages (from sympy==1.13.1->-r requirements.txt (line 15)) (1.3.0)
Requirement already satisfied: py-cpuinfo in /usr/local/lib/python3.11/dist-packages (from ultralytics>=8.2.34->-r requirements.txt (line 18)) (9.0.0)
Collecting ultralytics-thop>=2.0.0 (from ultralytics>=8.2.34->-r requirements.txt (line 18))
  Downloading ultralytics_thop-2.0.14-py3-none-any.whl.metadata (9.4 kB)
Requirement already satisfied: nvtx>=0.2.0 in /usr/local/lib/python3.11/dist-packages (from pandas>=1.1.4->-r requirements.txt (line 27)) (0.2.9)
```

```
from google.colab import files
uploaded = files.upload()
```

Choose Files No file chosen

Upload widget is only available when the cell has been executed in the current browser session. Please rerun this cell to

```
import os
from pathlib import Path
img_path = list(uploaded.keys())[0]
!python detect.py --weights yolov5s.pt --img 640 --conf 0.25 --source {img_path}
```

↗ Creating new Ultralytics Settings v0.0.6 file ☒
View Ultralytics Settings with 'yolo settings' or at '/root/.config/Ultralytics/settings.json'
Update Settings with 'yolo settings key=value', i.e. 'yolo settings runs_dir=path/to/dir'. For help see <https://docs.ultralytics.com/quickstart/#ultralytics-settings>
detect: weights=['yolov5s.pt'], source=kitty-cat-kitten-pet-45201.jpeg, data=data/coco128.yaml, imgsz=[640, 640], conf_thres=0.25, iou_t
YOLOv5 🚀 v7.0-419-gcd44191c Python-3.11.12 torch-2.6.0+cu124 CPU

Downloading <https://github.com/ultralytics/yolov5/releases/download/v7.0/yolov5s.pt> to yolov5s.pt...
100% 14.1M/14.1M [00:00<00:00, 210MB/s]

Fusing layers...
YOLOv5s summary: 213 layers, 7225885 parameters, 0 gradients, 16.4 GFLOPs
image 1/1 /content/yolov5/kitty-cat-kitten-pet-45201.jpeg: 640x640 1 cat, 546.1ms
Speed: 3.6ms pre-process, 546.1ms inference, 33.1ms NMS per image at shape (1, 3, 640, 640)
Results saved to runs/detect/exp

```
from IPython.display import Image, display
result_img_path = Path("runs/detect/exp") / img_path
display(Image(filename = result_img_path))
```



